Residual Gas Velocity Above a Stopped Rotating Disk in Dense Xenon: A Mechanism-by-Mechanism Analysis

1 Problem Definition

A solid horizontal disk of radius R rotates inside a vertical cylindrical vessel filled with xenon gas at high pressure. The disk sweeps through a prescribed angle θ_{sweep} using bang-bang kinematics (maximum acceleration then maximum deceleration), starting from rest and returning to rest. We seek the residual gas velocity at a height z_{obs} above the disk surface, evaluated at a time t_{obs} after the disk has stopped.

Unlike the propeller geometry (isolated NACA 0012 blades), the disk presents a *continuous solid surface* of area πR^2 to the gas above. The boundary layer is a von Kármán rotating-disk boundary layer, which is qualitatively different from a flat-plate airfoil BL.

1.1 Parameters

Seven parameters are treated as variable inputs; all others are fixed.

Table 1: Variable parameters (with default values).

Symbol	Description	Default	Units
h_{disk}	Disk thickness	0.005	m
θ_{sweep}	Sweep angle	$\pi/8$	rad
m_{disk}	Mass of disk	50	kg
m_{plate}	Mass of each carrier plate	0.15	kg
τ	Motor torque	60	N·m
z_{obs}	Observation height above disk	0.005	m
t_{obs}	Observation time after stop	0.5	s

Table 2: Fixed parameters.

Parameter	Value	Description
R	1.6 m	Disk radius
P	15 bar	Xenon pressure
T	300 K	Gas temperature
ΔD	0.010 m	Vessel inner diameter surplus
H	1.5 m	Vessel height
N	8	Number of carrier positions

1.2 Geometry and Coordinate System

The disk rotates in the xy-plane with its upper surface at $z = h_{\text{disk}}$ above the vessel floor. The z-axis points upward. The cylindrical vessel has inner radius

$$R_{\text{cyl}} = R + \frac{\Delta D}{2}\,, \tag{1}$$

giving a radial gap $\Delta r = \Delta D/2 = 5\,\text{mm}$. The gas column above the disk has height $H_{\text{gas}} = H - h_{\text{disk}} \approx 1.495\,\text{m}$.

2 Xenon Gas Properties

Identical to the propeller analysis. At $P = 15\,\text{bar}$, $T = 300\,\text{K}$:

$$\rho = \frac{PM}{R_g T} \qquad \approx 79\,\text{kg/m}^3\,, \tag{2}$$

$$\mu \qquad \approx 2.32 \times 10^{-5}\,\text{Pa}\cdot\text{s}\,, \tag{3}$$

$$\nu = \mu/\rho \qquad \approx 2.94 \times 10^{-7}\,\text{m}^2/\text{s}\,, \tag{4}$$

$$c_s = \sqrt{\gamma R_g T/M} \approx 178\,\text{m/s}\,. \tag{5}$$

The key feature: ν is *very* small ($51\times$ smaller than air at STP), leading to high Reynolds numbers at modest velocities and extremely slow molecular diffusion.

3 Kinematics

3.1 Moment of Inertia

The disk is a uniform solid cylinder of mass m_{disk} and radius R:

$$I_{\text{disk}} = \tfrac{1}{2}\, m_{\text{disk}}\, R^2\,. \tag{6}$$

The N carrier plates are housed in the hangar at $r \approx R$:

$$I_{\text{plates}} = N\, m_{\text{plate}}\, R^2\,. \tag{7}$$

Total:

$$I = I_{\text{disk}} + I_{\text{plates}} = \tfrac{1}{2}\, m_{\text{disk}}\, R^2 + N\, m_{\text{plate}}\, R^2\,. \tag{8}$$

For default parameters ($m_{\text{disk}} = 50\,\text{kg}$, $m_{\text{plate}} = 0.15\,\text{kg}$, $N = 8$): $I_{\text{disk}} = 64.0\,\text{kg}\cdot\text{m}^2$, $I_{\text{plates}} = 3.07\,\text{kg}\cdot\text{m}^2$, $I = 67.1\,\text{kg}\cdot\text{m}^2$.

3.2 Aerodynamic Drag Torque

A rotating disk in a fluid experiences a resisting torque from skin friction on both the upper and lower surfaces. For a turbulent boundary layer on one side of the disk, the moment coefficient is (Schlichting & Gersten, *Boundary-Layer Theory*, 9th ed., Ch. 5):

$$C_M = \frac{0.146}{\text{Re}_R^{1/5}}\,, \tag{9}$$

where $\text{Re}_R = \omega R^2/\nu$ is the rotational Reynolds number. The torque on one side is

$$\tau_{\text{one side}} = \tfrac{1}{2}\, C_M\, \rho\, \omega^2\, R^5\,. \tag{10}$$

For two sides (upper and lower):

$$\tau_{\text{drag}} = C_M \, \rho \, \omega^2 \, R^5 \, . \tag{11}$$

Note that C_M depends on ω through Re_R, making the drag nonlinear. For the edge contribution (rim drag), we add a bluff-body drag on the annular edge of thickness h_{disk}:

$$\tau_{\text{rim}} = \tfrac{1}{2} \, \rho \, C_{d,\text{rim}} \, h_{\text{disk}} \, R^4 \, \omega^2 \, , \tag{12}$$

with $C_{d,\text{rim}} \approx 1.0$ (blunt edge, Hoerner, *Fluid-Dynamic Drag*, Ch. 3).

3.3 Bang-Bang Motion

The equation of motion is

$$I \, \frac{d\omega}{dt} = \pm \tau_{\text{motor}} - \tau_{\text{drag}}(\omega) - \tau_{\text{rim}}(\omega) \, , \tag{13}$$

where the sign alternates between acceleration and deceleration phases. The switching angle θ_{switch} is found numerically such that $\omega \to 0$ exactly when $\theta \to \theta_{\text{sweep}}$.

4 Boundary Layer: Von Kármán Rotating Disk

4.1 The Classical Problem

The boundary layer on an infinite rotating disk in an otherwise quiescent fluid was solved by von Kármán (1921) and refined by Cochran (1934). The key result: the BL thickness is *independent of radius* and depends only on rotation rate and viscosity:

$$\delta = a \sqrt{\frac{\nu}{\omega}} \, , \tag{14}$$

where $a \approx 5.5$ (laminar, Schlichting & Gersten, Ch. 5.2). The BL has three velocity components:

- **Tangential** (v_ϕ): the gas near the disk is dragged in the direction of rotation.
- **Radial** (v_r): centrifugal force pushes the gas outward within the BL.
- **Axial** (v_z): to replace the gas expelled radially, fresh gas is drawn downward from above the BL. This is the *von Kármán pumping* effect.

The axial (pumping) velocity far from the disk is (von Kármán, 1921; Cochran, 1934):

$$w_\infty = -0.886 \sqrt{\nu \, \omega} \qquad \text{(laminar)} \, . \tag{15}$$

The negative sign indicates flow *toward* the disk. This means the rotating disk *sucks gas down from above* uniformly across its entire surface.

4.2 Turbulent Regime

The rotating-disk BL transitions to turbulence at $\text{Re}_r = \omega r^2/\nu \approx 2.5 \times 10^5$ (Lingwood, 1995). In the turbulent regime, the BL thickness grows:

$$\delta_{\text{turb}} \approx 0.526 \, r \left(\frac{\nu}{\omega r^2} \right)^{1/5} = 0.526 \, r \, \text{Re}_r^{-1/5} \, , \tag{16}$$

(Owen & Rogers, *Flow and Heat Transfer in Rotating-Disc Systems*, Vol. 1, 1989). Unlike the laminar case, the turbulent BL thickness *increases with radius*.

The turbulent axial pumping velocity also increases:

$$w_\infty^{\rm turb} \approx -C_w \sqrt{\nu_t\,\omega}\,, \tag{17}$$

where the turbulent viscosity $\nu_t \gg \nu$. A more practical estimate uses the entrainment flow rate. The volumetric flow entrained by one side of the disk (turbulent) is (Owen & Rogers, 1989):

$$Q = 0.219\,{\rm Re}_R^{4/5}\,\nu\,R\,, \tag{18}$$

where ${\rm Re}_R = \omega R^2/\nu$. The mean axial velocity at the disk edge (over the disk area):

$$\bar{w}_{\rm axial} = \frac{Q}{\pi R^2}\,. \tag{19}$$

4.3 Relevance to the MARS Problem

This is the fundamental difference from the propeller: a rotating disk has an *intrinsic axial pumping velocity* that draws gas *toward* the disk surface. During rotation, this sets up a meridional circulation (radially outward in the BL, axially downward above the disk). After the disk stops, this circulation must decay.

The question is: how quickly does the axial flow decay, and what is its magnitude at $z = z_{\rm obs}$ at time $t = t_{\rm obs}$?

5 Mechanism 1: Von Kármán Pumping and Decay

5.1 Axial Velocity During Rotation

During steady rotation at angular velocity ω, the axial (pumping) velocity depends on height above the disk. In the laminar regime (Cochran's solution):

$$w(z) = -\sqrt{\nu\omega}\;H_{\rm vK}(\zeta)\,, \qquad \zeta = z\sqrt{\omega/\nu}\,, \tag{20}$$

where $H_{\rm vK}(\zeta)$ is a universal function that increases from 0 at $\zeta = 0$ (disk surface, no-penetration) to 0.886 as $\zeta \to \infty$. The asymptotic value is reached for $\zeta \gtrsim 5$ (i.e. $z \gtrsim \delta$).

For the turbulent regime at the outer part of the disk, the pumping is stronger. The mean downward velocity can be estimated from the entrainment flow rate (18).

5.2 Decay After Stopping

When the disk stops at $t = 0$, the forcing that sustains the BL and the pumping vanishes instantaneously. The decay proceeds through several stages:

Stage 1: Turbulent decay ($0 < t \lesssim 2\,t_e$). The turbulent eddies in the BL lose their energy source and decay on the eddy turnover timescale:

$$t_e = \frac{\delta}{\omega\,r\cdot 0.05}\,, \tag{21}$$

where $0.05\,\omega r$ is the turbulence intensity. The turbulent viscosity decays exponentially with e-folding time $\sim 2\,t_e$. During this period the BL profile changes from a turbulent 1/7-power law to a lower-energy state.

Stage 2: Molecular spin-down ($t \gg 2\,t_e$). After the turbulence dies, only molecular viscosity remains. The tangential and radial velocity components diffuse outward from the disk surface. The tangential velocity profile evolves as

$$v_\phi(z,t) \sim V_\phi(0)\,\mathrm{erfc}\left(\frac{z}{2\sqrt{\nu t}}\right), \tag{22}$$

where $V_\phi(0)$ is the initial tangential velocity at the edge of the decayed BL. This is the classical Stokes first problem (Schlichting & Gersten, Ch. 4.3).

Axial velocity. The axial pumping is *driven by* the tangential and radial velocities in the BL via the continuity equation. Once the tangential velocity decays (no more centrifugal force in the BL), the radial outflow ceases, and the axial inflow ceases with it. The axial velocity decays on the *same timescale* as the tangential BL.

The axial pumping velocity scales as $w \sim \sqrt{\nu_{\mathrm{eff}}\,\omega_{\mathrm{eff}}}$ where ω_{eff} is the effective angular velocity of the gas near the disk surface. As $\omega_{\mathrm{eff}} \to 0$, $w \to 0$.

5.3 Quantitative Estimate

During rotation (peak values). The peak angular velocity $\omega_{\max}$ is obtained from the bang-bang kinematics. The laminar pumping velocity:

$$w_{\mathrm{lam}} = 0.886\,\sqrt{\nu\,\omega_{\max}}\,. \tag{23}$$

The turbulent entrainment flow rate gives the mean axial velocity over the disk area. This represents the steady-state value during rotation.

After stop. The PDE governing the tangential velocity is

$$\frac{\partial v_\phi}{\partial t} = \frac{\partial}{\partial z}\left[\nu_{\mathrm{eff}}(z,t)\,\frac{\partial v_\phi}{\partial z}\right], \tag{24}$$

with $v_\phi(0,t) = 0$ (no-slip on stopped disk), $v_\phi(z \to \infty, t) = 0$, and initial condition from the von Kármán profile. This is essentially the same 1D diffusion equation as for the propeller, but with a different initial condition and with the crucial difference that the initial profile extends over the entire disk area, not just N narrow blade wakes.

The axial velocity at z_{obs} is then obtained from continuity:

$$w(z_{\mathrm{obs}},t) \approx -\frac{\partial}{\partial r}\int_0^{z_{\mathrm{obs}}} v_r(r,z,t)\,dz\,. \tag{25}$$

In practice, for $z_{\mathrm{obs}} > \delta$, the axial velocity is estimated as $w \sim \sqrt{\nu_{\mathrm{eff}}(t)\cdot\omega_{\mathrm{eff}}(t)}$, which decays as both ν_t and ω_{eff} decay.

5.4 Result

The von Kármán pumping produces a nonzero axial velocity *during* rotation. After the disk stops, this axial velocity decays on the same timescale as the tangential BL. For $t_{\mathrm{obs}} \gg 2\,t_e$, the pumping has ceased. However, the magnitude and decay time depend strongly on the disk angular velocity. The companion Python script evaluates this numerically.

$$\boxed{u(z_{\mathrm{obs}},\,t_{\mathrm{obs}}) = \text{computed numerically; see Python script}} \tag{26}$$

6 Mechanism 2: Tangential BL Diffusion

Even without the axial pumping, the tangential (azimuthal) velocity in the BL can in principle diffuse vertically, exactly as in the propeller case.

6.1 Initial Tangential Profile

The von Kármán tangential velocity profile (turbulent, 1/7-power law approximation):

$$v_\phi(z) = \omega r \left[1 - \left(\frac{z}{\delta}\right)^{1/7}\right] \qquad (0 < z \leq \delta)\,, \tag{27}$$

where δ is the local turbulent BL thickness. At the tip ($r = R$), $v_\phi(0) = \omega R = V_{\text{tip}}$.

6.2 Diffusion Equation

Same 1D diffusion equation as for the propeller:

$$\frac{\partial v_\phi}{\partial t} = \frac{\partial}{\partial z}\left[\nu_{\text{eff}}(z,t)\,\frac{\partial v_\phi}{\partial z}\right], \tag{28}$$

with initial condition from the von Kármán profile and decaying ν_t.

6.3 Key Difference from Propeller

The propeller BL has thickness $\delta = 0.37\,c\,\text{Re}_c^{-1/5}$, which depends on the chord c. The disk BL thickness at radius r is $\delta = 0.526\,r\,\text{Re}_r^{-1/5}$, which depends on *radius*. At the outer edge ($r = R = 1.6\,\text{m}$), the disk BL is much thicker than a propeller blade BL (because $R \gg c$).

Whether $z_{\text{obs}} - \delta > 0$ (observation above the BL) or $z_{\text{obs}} - \delta < 0$ (observation inside the BL) depends on the disk angular velocity and observation height.

6.4 Result

The 1D PDE solver evaluates the tangential velocity at z_{obs}, t_{obs}. If $z_{\text{obs}} > \delta + \ell_{\text{turb}} + \ell_{\text{mol}}$, the result is zero. If the observation point is close to δ, a small nonzero tail may exist (same as propeller).

$$\boxed{v_\phi(z_{\text{obs}},\, t_{\text{obs}}) = \text{computed numerically; see Python script}} \tag{29}$$

7 Mechanism 3: Bulk Swirl / Ekman Spin-Down

7.1 Angular Impulse

The disk drag torque deposits angular momentum in the gas. Unlike the propeller (where the drag acts on N narrow blades), the disk torque acts over the entire disk surface:

$$\tau_{\text{drag}} = C_M\,\rho\,\omega^2\,R^5 + \tfrac{1}{2}\,\rho\,C_{d,\text{rim}}\,h_{\text{disk}}\,R^4\,\omega^2\,. \tag{30}$$

The angular impulse over the sweep is $J = \int_0^{T_{\text{sweep}}} \tau_{\text{drag}}\,dt$.

7.2 Ekman Spin-Down

In a confined vessel, the bulk swirl decays via the Ekman mechanism: endcap boundary layers transport angular momentum to the walls. The Ekman spin-down time is

$$t_{\text{Ekman}} = \frac{H}{\sqrt{\nu\,\Omega_{\text{gas}}}}\,, \tag{31}$$

where Ω_{gas} is the characteristic gas rotation rate. For molecular ν, this is $\sim 1000\,\text{s}$; for turbulent ν_t (lasting $\sim 10\,\text{ms}$), $\sim 50\,\text{s}$.

The Ekman pumping velocity $w_E = \sqrt{\nu\,\Omega}$ is of order $\sim 1\,\text{mm/s}$ (molecular), producing negligible vertical displacement in t_{obs}.

7.3 Result

$$\boxed{u(z_{\text{obs}},\, t_{\text{obs}}) \approx 0 \quad \text{(bulk Ekman spin-down)}} \tag{32}$$

Same conclusion as for the propeller: the Ekman mechanism is far too slow to affect the observation.

8 Mechanism 4: Displacement (Potential) Flow

As the disk rotates, the rim (a blunt edge of height h_{disk}) pushes gas aside. The displacement flow around the rim produces a transient velocity field that decays at the speed of sound after the disk stops.

8.1 Clearing Time

$$t_{\text{acoustic}} = \frac{R}{c_s} \approx \frac{1.6\,\text{m}}{178\,\text{m/s}} \approx 9\,\text{ms}\,. \tag{33}$$

8.2 Blockage

The disk blockage ratio is $h_{\text{disk}}/\Delta r = 1.0$ (for default parameters), much smaller than the propeller blade blockage (5.3). The displacement flow is correspondingly weaker.

8.3 Result

$$\boxed{u(z_{\text{obs}},\, t_{\text{obs}}) = 0 \quad \text{(displacement flow, for } t_{\text{obs}} \gg t_{\text{acoustic}})} \tag{34}$$

9 Mechanism 5: Pressure Pulse from Deceleration

Same physics as for the propeller. The deceleration of the disk creates a transient pressure perturbation:

$$\Delta p \sim \rho\, V_{\text{tip}}^2\,, \tag{35}$$

which propagates as an acoustic wave passing through z_{obs} in $\sim z_{\text{obs}}/c_s$ and leaving no residual.

$$\boxed{u(z_{\text{obs}},\, t_{\text{obs}}) = 0 \quad \text{(pressure pulse, for } t_{\text{obs}} \gg t_{\text{decel}})} \tag{36}$$

10 Mechanism 6: Rim Separation and Wake

The disk has a blunt rim of height h_{disk}, which produces flow separation at the edge during rotation. Unlike the streamlined NACA 0012 trailing edge of the propeller, the disk edge generates a separated wake.

10.1 Wake Character

The separated wake at the rim is *tangential*: it lies in the plane of the disk edge, in the annular gap between the disk and the vessel wall. The wake has a vertical extent of order $\sim h_{\text{disk}} = 5\,\text{mm}$, and its velocity decays after the disk stops.

10.2 Vertical Transport

The rim wake is tangential and confined to the disk-edge plane. It does not have an axial velocity component directed upward toward z_{obs}. The same transport limitations as for the propeller vortex shedding apply: vertical diffusion from the rim wake is limited by the same BL-thickness-plus-diffusion reach constraint.

10.3 Result

$$\boxed{u(z_{\text{obs}},\, t_{\text{obs}}) = 0 \quad \text{(rim wake, confined to edge)}} \tag{37}$$

11 Mechanism 7: Secondary Flows (Centrifugal, Coriolis)

11.1 Centrifugal Effects in the Bulk

Gas in the bulk (above the BL) that has acquired angular momentum is subject to centrifugal force. This drives radial outflow, which must be compensated by axial inflow from above (this is the von Kármán pumping, already treated in Section 5).

After the disk stops, the centrifugal force decays with the angular velocity of the gas. No new mechanism beyond what is already accounted for in Section 5.

11.2 Rim-Gap Couette Flow

In the radial clearance Δr between the disk edge and the vessel wall:

$$\text{Re}_{\text{gap}} = \frac{V_{\text{tip}}\,\Delta r}{\nu}\,. \tag{38}$$

This flow is purely tangential (azimuthal shear between moving disk edge and stationary wall), with no axial velocity component. Same conclusion as for the propeller tip gap.

11.3 Result

$$\boxed{u(z_{\text{obs}},\, t_{\text{obs}}) = 0 \quad \text{(secondary flows)}} \tag{39}$$

12 Vessel Confinement Effects

The same confinement analysis as for the propeller applies. All five effects (wall BLs, Ekman spin-up, tip-gap Couette, reflected pressure waves, displacement readjustment) are quantified in the companion Python script and shown to be negligible. See the propeller analysis (`propeller_flow_analysis.tex`, Section 7) for the detailed treatment; the physics is identical.

13 Key Difference from Propeller: Axial Pumping

The propeller analysis found that *all* mechanisms give zero at z_{obs} (provided $z_{\text{obs}} - \delta >\sim 2\,\text{mm}$). For the rotating disk, the situation is qualitatively different because of the **von Kármán pumping**:

- The propeller blades have no axial flow: at $\alpha = 0$, the NACA 0012 produces no lift and no axial pressure gradient. All flow is tangential.
- The rotating disk intrinsically generates an axial velocity ($w_\infty = -0.886\sqrt{\nu\omega}$ in the laminar regime, stronger in the turbulent regime). This is a *volume effect*: the entire disk area participates.
- The pumping extends to $z \to \infty$ (in an infinite domain), meaning it reaches z_{obs} during rotation.
- After the disk stops, the pumping decays as the BL tangential velocity decays. The question is whether this decay is fast enough.

The BL thickness is also much larger for the disk than for the propeller. At the disk edge ($r = R = 1.6\,\text{m}$):

$$\delta_{\text{disk}}(R) = 0.526\, R\, \text{Re}_R^{-1/5}\,, \tag{40}$$

compared with the propeller:

$$\delta_{\text{prop}} = 0.37\, c\, \text{Re}_c^{-1/5}\,. \tag{41}$$

Since $R = 1.6\,\text{m} \gg c = 0.22\,\text{m}$, the disk BL is significantly thicker, potentially engulfing z_{obs}.

Both quantities (pumping velocity and BL thickness) are evaluated numerically in the companion Python script.

14 Conclusion

Table 3 summarises all mechanisms.

Table 3: Summary of all mechanisms for residual velocity at z_{obs} above the disk, t_{obs} after stopping.

Mechanism	Characteristic timescale / Comment	$u(z_{\text{obs}}, t_{\text{obs}})$
Von Kármán pumping (axial)	Decays with BL tangential velocity	Numerical
Tangential BL diffusion	Same as propeller eddy diffusion	Numerical
Bulk Ekman spin-down	$t_{\text{Ekman}} \sim 1000\,\text{s}$ (mol.)	0
Displacement flow	$t_{\text{acoustic}} \approx 9\,\text{ms}$	0
Pressure pulse	$t_{\text{decel}} \sim 10\,\text{ms}$	0
Rim wake	Confined to disk edge	0
Secondary flows	Centrifugal decays with ω; gap flow tangential	0

The critical difference from the propeller is the **von Kármán pumping**, which provides a direct axial velocity mechanism that extends above the disk BL. The propeller has no such mechanism. Whether this makes the disk geometry worse depends on the decay rate of the pumping after the disk stops, which is evaluated quantitatively in the companion script `disk_flow_numerics.py`.

A second concern is the **BL thickness**: the disk BL at $r = R$ may be thick enough to engulf z_{obs}, in which case the observation point is inside the BL and the velocity is necessarily nonzero. The propeller BL (chord-based) is thinner.